





$\alpha$

$\beta$

$\mu_\alpha, \Sigma_\alpha$

$\mu_\beta, \Sigma_\beta$





using only first two moments





cheap surrogate



# Gelbrich Bound on $W_2$

For **arbitrary** measures  $\alpha$  and  $\beta$  with means/covars  $\mu_\alpha, \Sigma_\alpha$  and  $\mu_\beta, \Sigma_\beta$

$$\|\mu_\alpha - \mu_\beta\|^2 + \mathcal{B}(\Sigma_\alpha, \Sigma_\beta)^2 \leq W_2^2(\alpha, \beta)$$

Interpretation:

- Lower bound on Wasserstein-2 **using only first two moments**.
- Tight (equality) when both measures are Gaussian.

Applications:

- Useful as a **cheap surrogate** for  $W_2$  in ML
- Used in domain adaptation, distributional robustness, variational inference.

# $W_2$ factors out translations

Consider the translation operator  $T_\tau : x \mapsto x - \tau$ . One has:

$$W_2(T_{\tau\#}\alpha, T_{\tau'\#}\beta)^2 = W_2(\alpha, \beta)^2 - 2\langle \tau - \tau', \mu_\alpha - \mu_\beta \rangle + \|\tau - \tau'\|^2$$